Unsupervised Learning & Representation

Prof Yueming Xing

What Is Unsupervised Learning?

FOUNDATIONS

The Core Problem

We observe inputs from an unknown distribution, but no labels. In supervised learning, we model $P(Y \mid X)$; here, we model $P(X)$.

This is harder: there is no clear answer, and evaluation is less direct.

Why It Matters

Labeled data is expensive and limited. Most real-world data is unlabeled, so unsupervised methods reveal structure and patterns.

Understanding the data distribution is essential before specific algorithms.

No Labels

Only inputs X are available.

Model P(X)

We model the data distribution directly.

Ambiguous Success

Without ground truth, evaluation changes.

Representation Learning

CORE CONCEPT

Unsupervised learning can be viewed through **representation learning**: turning raw inputs into a space where structure is clearer, noise is lower, and downstream tasks are easier.

Patterns Clearer

Distinct variation is easier to separate.

Noise Reduced

Irrelevant dimensions are compressed away.

Relationships Simplified

Downstream tasks become more data-efficient.

Types of Unsupervised Learning

TAXONOMY

Unsupervised learning is a family of methods for uncovering structure in data. At a graduate level, the key point is that **every method encodes assumptions** about geometry, density, independence, or normality.

Clustering

Partition data into groups of similar observations.

Density Estimation

Model the underlying probability distribution P(X).

Representation Learning

Learn a compact, structured encoding of inputs.

Anomaly Detection

Identify rare observations that deviate from normal structure.

K-Means Objective

CLUSTERING

K-means minimizes the **within-cluster sum of squared distances**, or variance within each cluster. It finds assignments and centroids that minimize squared Euclidean distance to each point's centroid.

This objective assumes **Euclidean geometry**, **approximately spherical** clusters, and **equally important** features. If those assumptions fail, it still converges, but the result may be meaningless.

Euclidean Distance

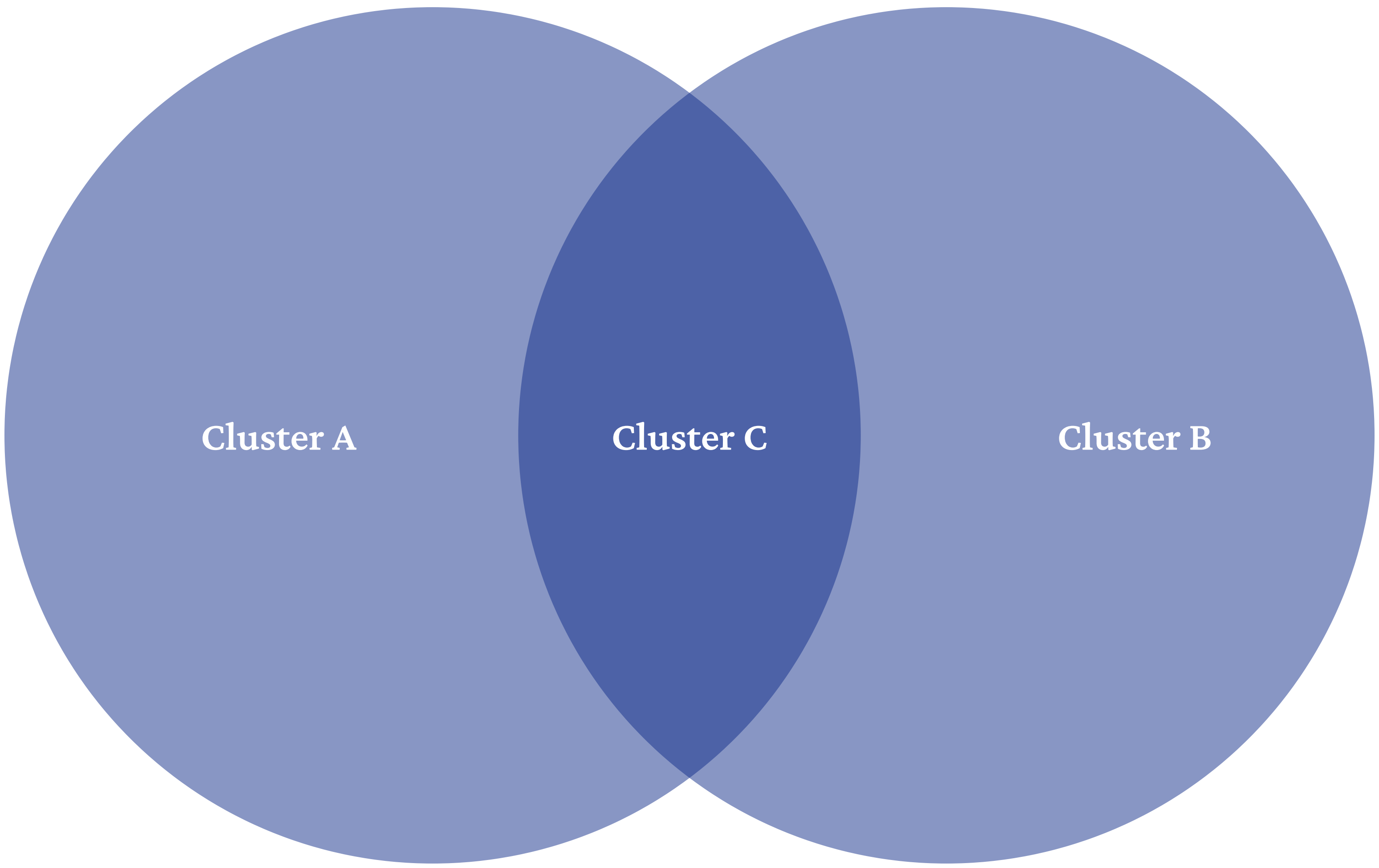
K-means uses squared Euclidean distance.

Spherical Clusters

Each group is summarized by one centroid.

Equal Feature Weight

All dimensions contribute uniformly.



Centroids represent assigned points, and the lines show the distances being minimized.

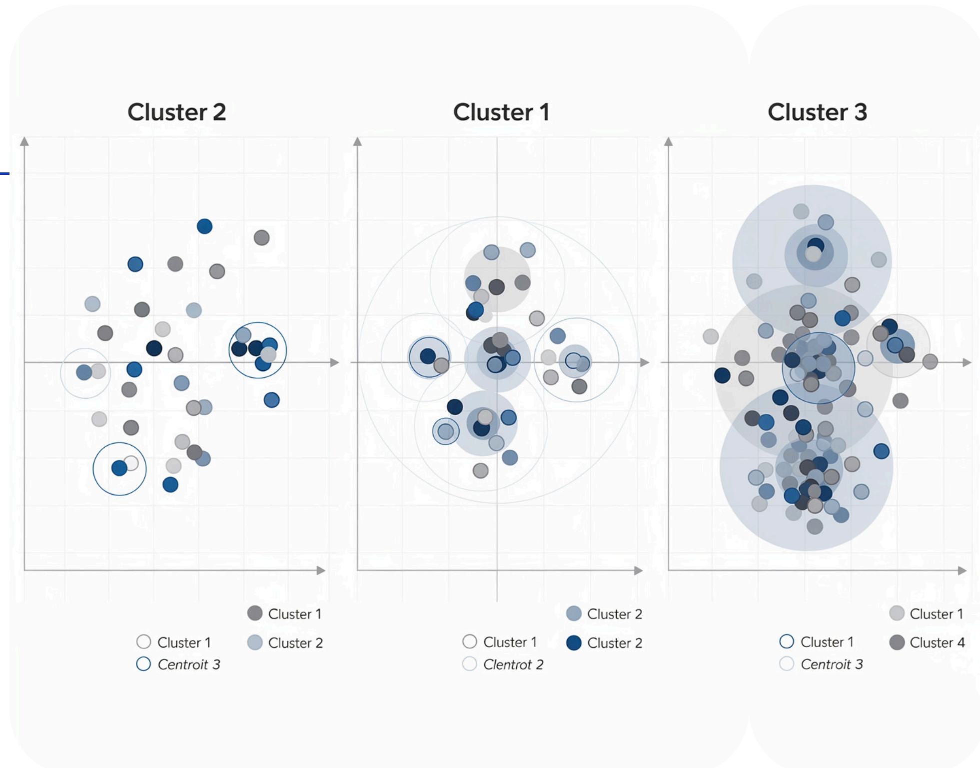
K-Means Algorithm

CLUSTERING

K-means alternates between **assignment** and **update**. Each iteration lowers the objective, so it converges, but only to a **local minimum** influenced by initialization.

Initial Centroids

Random centroids among scattered points with provisional boundaries



Updated Centroids

Centroids moved after reassignment with tighter cluster boundaries

Left: initial random centroids. Right: updated centroids after reassignment, with tighter clusters.

1

1. Initialize

Select k initial centroids randomly, by K-Means++, or otherwise.

2

2. Assign

Assign each point to its nearest centroid under Euclidean distance.

3

3. Update

Recompute each centroid as the mean of its assigned points.

4

4. Repeat

Continue until assignments stop changing and convergence is reached.

Limitations of K-Means

CRITICAL ANALYSIS

K-means struggles when its geometric assumptions are too rigid. These limits help explain why more flexible models are often needed.

Non-Spherical Clusters

K-means struggles with elongated, crescent-shaped, or interleaved clusters. Euclidean distance to a centroid is a poor fit for these shapes.

Sensitivity to Scaling

Large numeric ranges can dominate distance calculations. Scaling is a necessary preprocessing step, not an optional one.

Requires Specifying k

The number of clusters must be set before fitting. In practice, it is often estimated with heuristics like the elbow method or silhouette scores.

Local Minima

Poor initialization can trap K-means in a suboptimal partition. Multiple restarts or K-Means++ help reduce this risk.

Gaussian Mixture Models

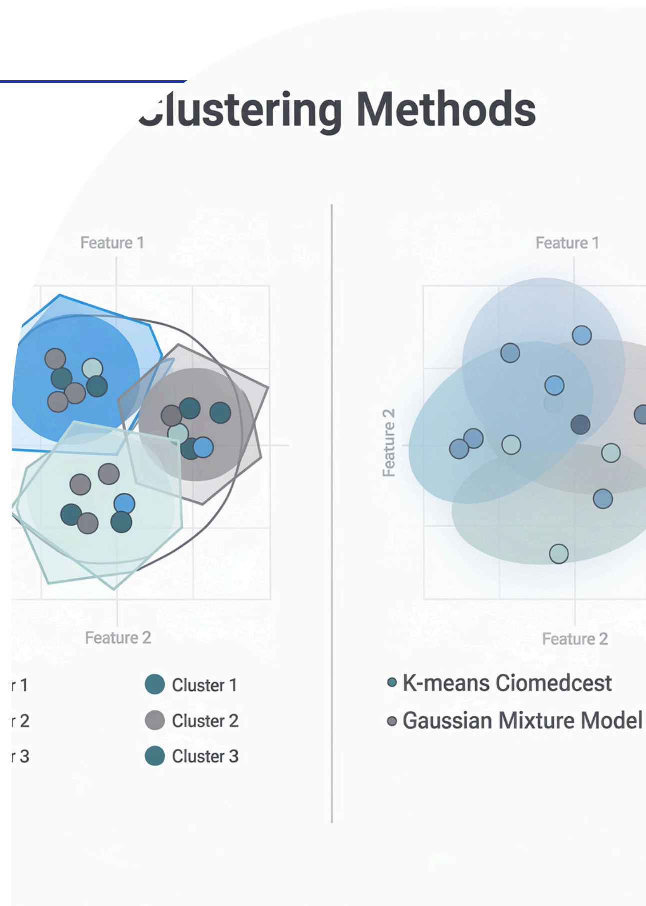
PROBABILISTIC CLUSTERING

Gaussian Mixture Models (GMMs) replace K-means' hard assignments with soft probabilities. Each point is modeled as coming from one of k Gaussian components, enabling elliptical clusters and boundary uncertainty.

K-Means	GMM
Hard assignments. Each point belongs to one cluster with spherical geometry.	Soft assignments. Each point has probabilities across clusters and can form elliptical shapes.

K-means

Hard boundaries with distinct colored regions.



GMM

Soft, overlapping Gaussian components with gradients.

Left: K-means uses hard Voronoi boundaries. Right: GMM uses overlapping probability ellipses, capturing uncertainty and non-spherical shapes.

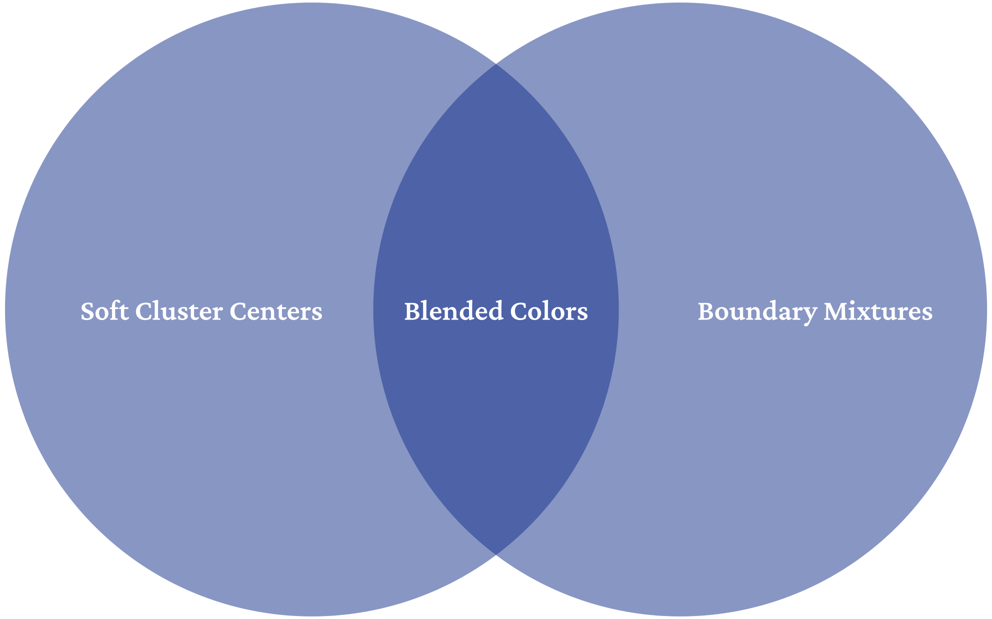
EM Algorithm — Intuition

OPTIMIZATION

The Core Challenge

We want to maximize the GMM's marginal likelihood, but each point's cluster assignment — the **latent variable** — is unknown. If we knew the assignments, optimization would be easy. If we knew the parameters, the assignments would be easy. We know neither.

This circular dependency makes direct optimization hard. EM breaks it by alternating between estimating soft assignments from the current parameters and updating the parameters from those estimates. It increases log-likelihood each iteration and converges to a local maximum.

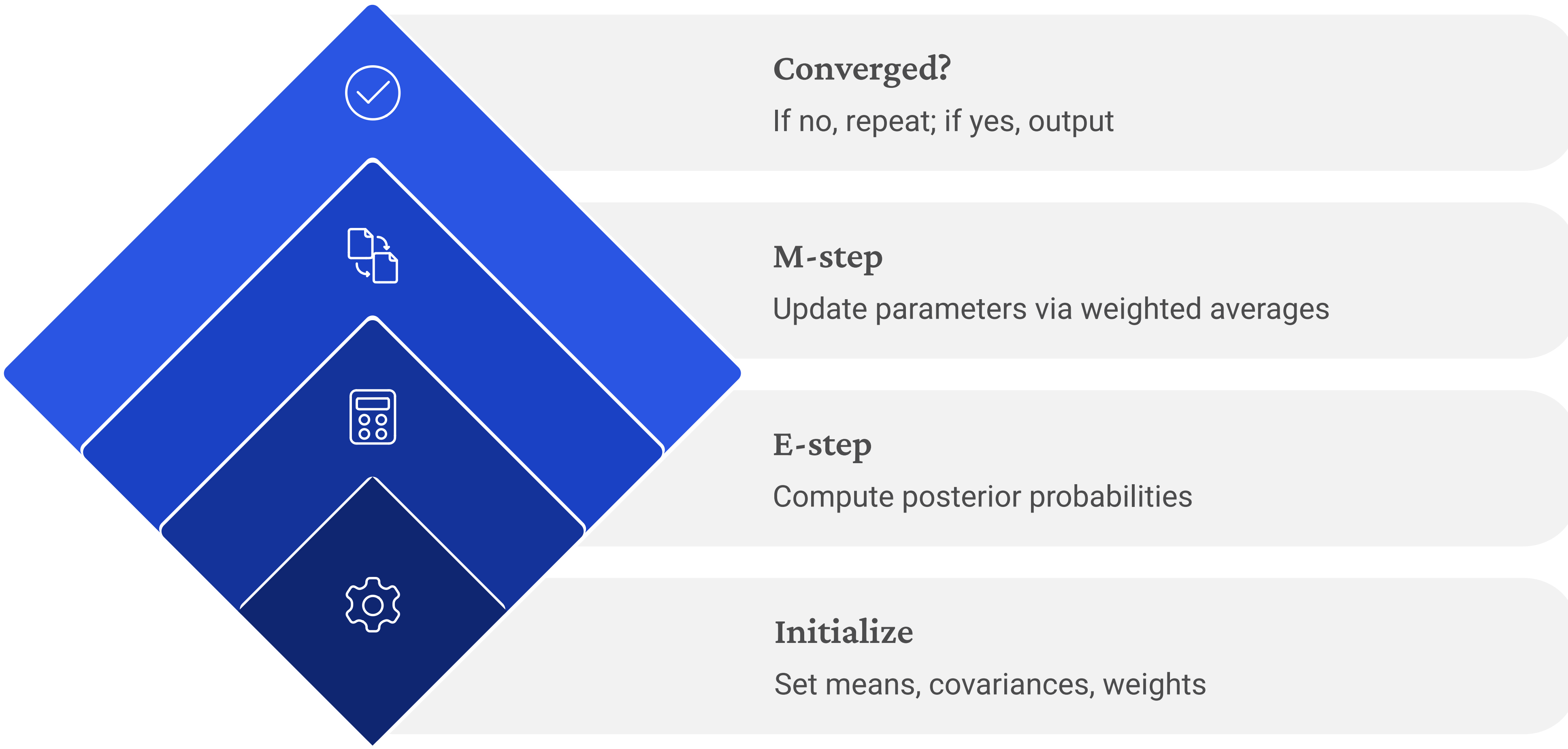


Points use soft memberships: pure colors for confident cases, mixed colors near boundaries.

EM Algorithm — Steps

OPTIMIZATION

The EM algorithm simplifies joint optimization into alternating steps. Each E-step plus M-step increases the marginal log-likelihood.



The E-step and M-step alternate until convergence. Each cycle increases the observed-data log-likelihood.

E-Step

Compute the posterior probability that each point belongs to each component. These soft responsibilities replace hard K-means assignments.

M-Step

Update means, covariances, and mixing proportions using the responsibilities as weights. Gaussian assumptions give closed-form updates.

Convergence

Repeat until the log-likelihood or parameters change less than a tolerance. Restarts can help avoid poor local maxima.

DBSCAN

DENSITY-BASED CLUSTERING

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) finds dense clusters in feature space, so it does not require k and marks sparse points as noise.

No k Required

Cluster count emerges from data density.

Arbitrary Shapes

Finds non-convex, elongated, and interleaved clusters.

Explicit Noise

Sparse points are labeled as noise.

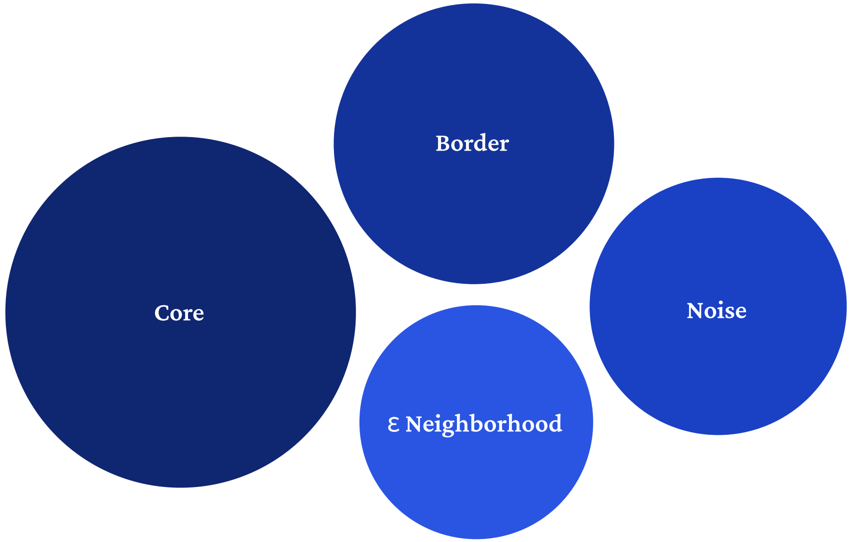
DBSCAN — Formal Idea

DENSITY-BASED CLUSTERING

Point Classification

DBSCAN uses two hyperparameters: **ε** (neighborhood radius) and **MinPts** (minimum points for a dense region). Every point falls into one of three categories.

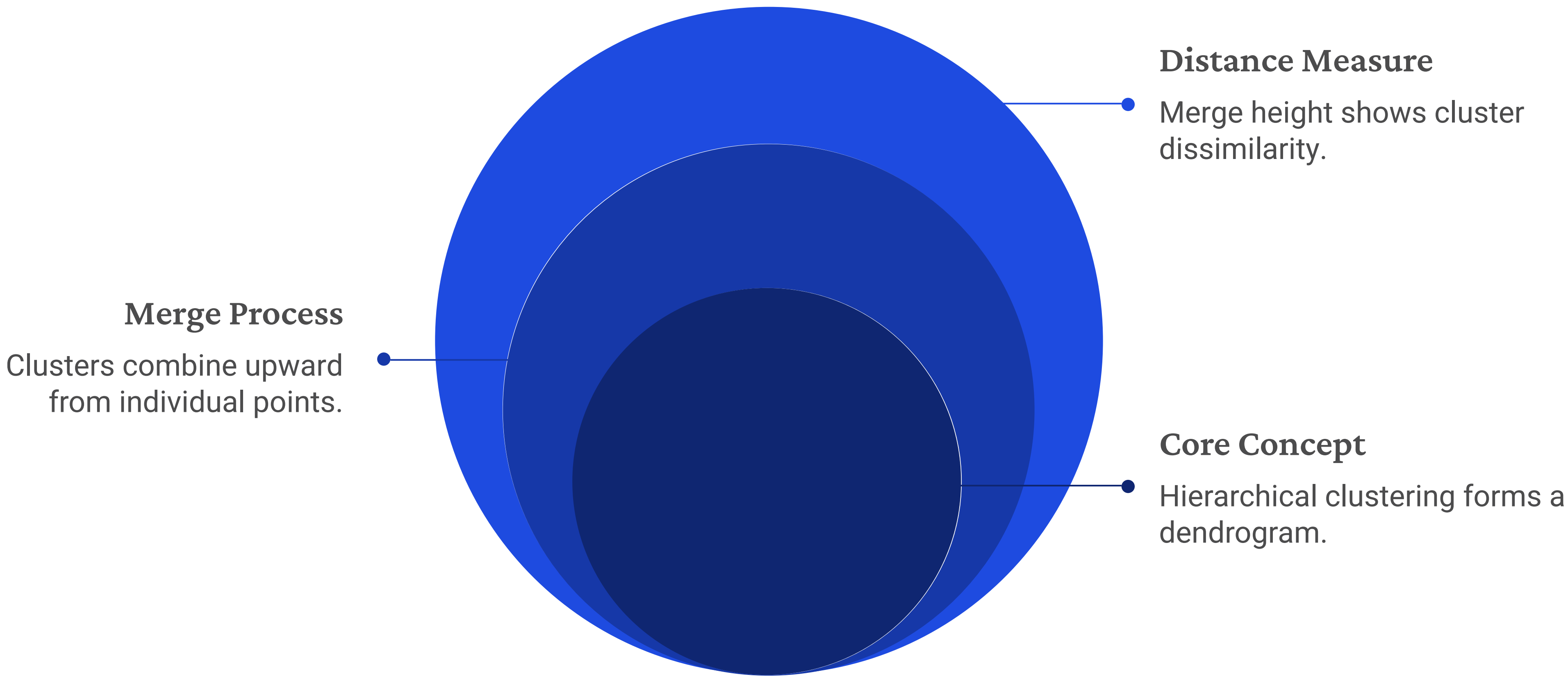
Core points have at least MinPts neighbors within ε. **Border points** have fewer neighbors but lie within ε of a core point. **Noise points** are neither core nor border, so they stay unclustered. Core points form clusters, while border points attach to nearby cores, enabling arbitrary shapes.



Hierarchical Clustering

HIERARCHICAL METHODS

Hierarchical clustering builds a **nested tree of partitions** called a dendrogram. It starts with each point as its own cluster, then merges the closest clusters step by step.



The dendrogram shows nested clusters, and a horizontal cut selects the final groups.

No k Required

You choose the cluster count by cutting the tree at a threshold.

Full Hierarchy

One dendrogram captures every level of cluster granularity.

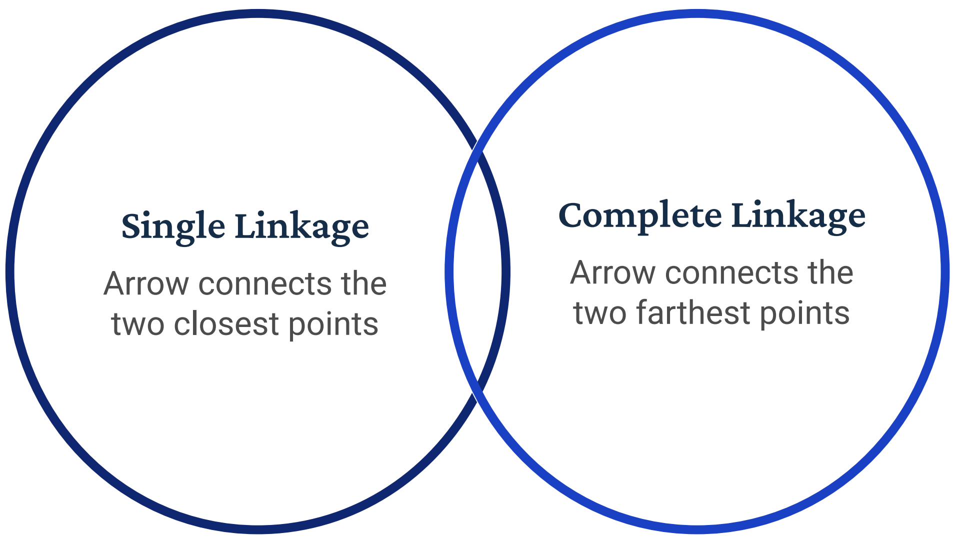
$O(n^2 \log n)$

It is more expensive than K-means and less practical for very large datasets.

Linkage Criteria

HIERARCHICAL METHODS

The linkage criterion defines how distance between two **groups** is computed from pairwise point distances. This choice can drastically change the cluster shapes and dendrograms produced.



Single linkage uses the minimum distance; complete linkage uses the maximum. Average linkage uses the mean, offering a middle ground.

Single Linkage

Minimum pairwise distance. Often forms elongated, chaining clusters, and can be influenced by bridging outliers.

Complete Linkage

Maximum pairwise distance. Produces compact clusters and is more robust to outliers.

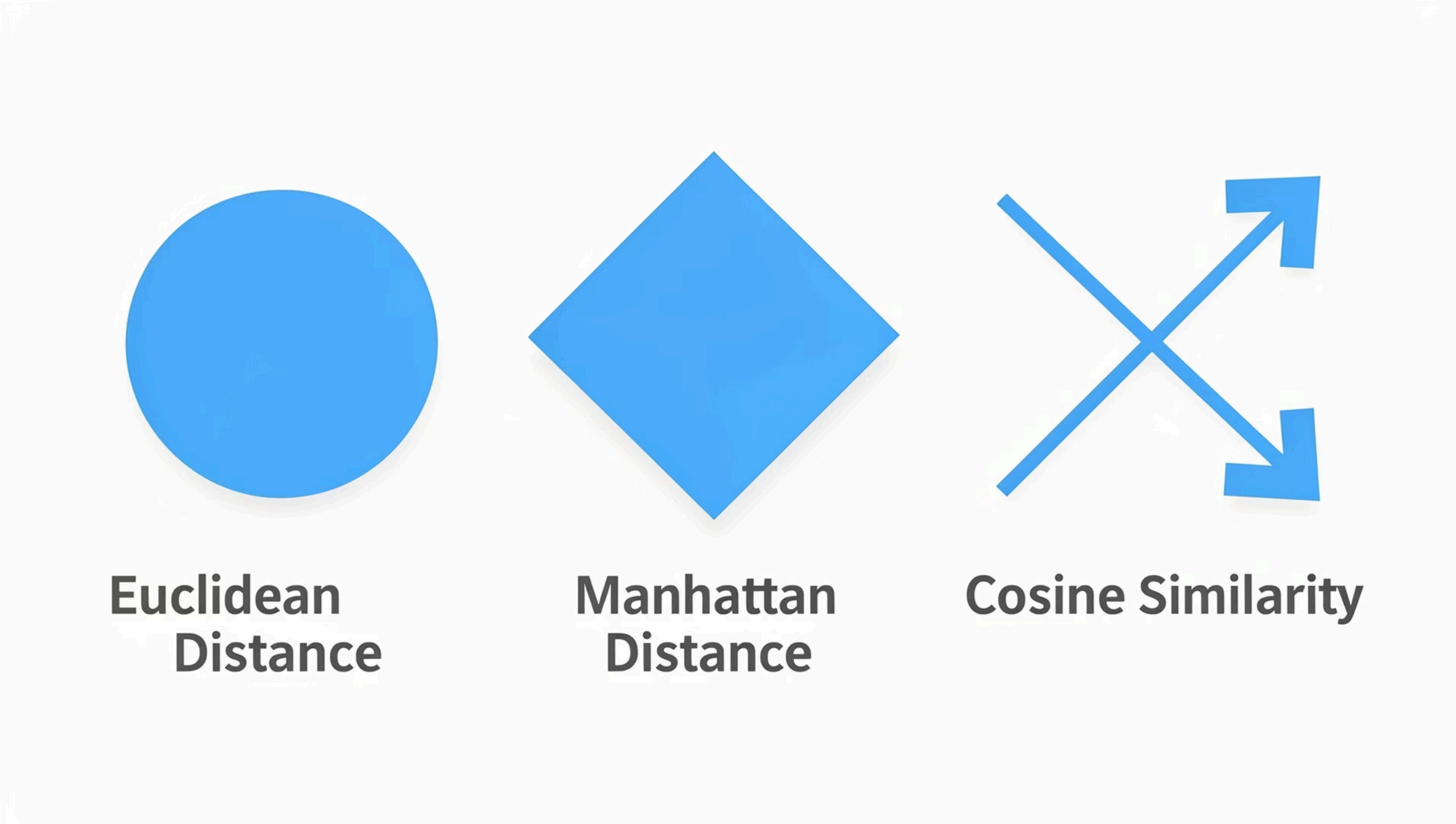
Average Linkage

Mean pairwise distance. A practical compromise with less chaining and less outlier sensitivity.

Distance Metrics

GEOMETRY OF SIMILARITY

Distance is the core primitive in clustering. The metric defines what “similar” means and shapes the geometry of the problem.



Each metric has a different notion of equal distance. Euclidean is spherical, Manhattan is diamond-shaped, and cosine ignores magnitude — making it useful for text and sparse data.

Euclidean

Straight-line distance. Best for continuous, interpretable features; used in K-means and GMM.

Manhattan

Sum of absolute differences. More robust to outliers and suited to grid-like spaces.

Cosine Similarity

Measures vector angle. Magnitude-invariant and ideal for text embeddings and sparse data.

Feature Scaling

PREPROCESSING

Feature scaling is essential for distance-based clustering. Without it, large-scale features dominate the distance calculation and hide the rest.

For example, salary differences can outweigh a binary gender indicator in Euclidean distance. The result reflects salary variance more than the full feature structure.

Standardization and min-max normalization both put features on comparable scales. The best choice depends on outliers and the downstream algorithm, but scaling is always a modeling decision.

Standardization (Z-score)

Subtract mean, divide by std. Maps to zero mean and unit variance. Preferred when features are approximately Gaussian.

Min-Max Normalization

Maps each feature to [0, 1]. Preserves relative spacing but is sensitive to outliers that compress the interior range.

Curse of Dimensionality

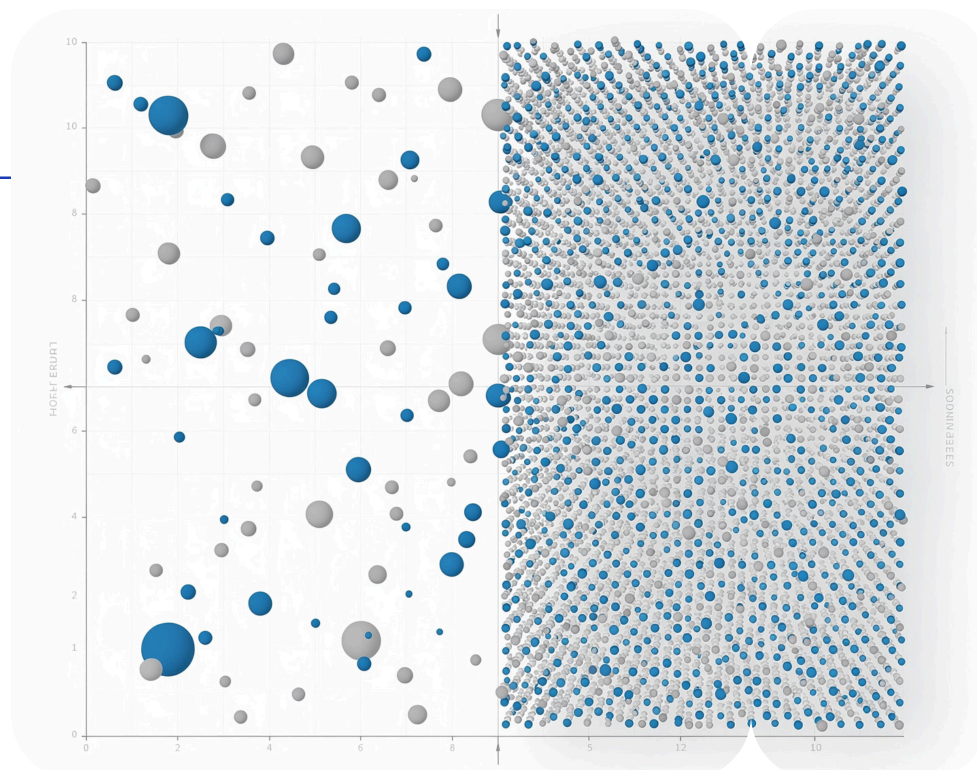
HIGH-DIMENSIONAL GEOMETRY

As features increase, data geometry changes in damaging ways. In high dimensions, **concentration of measure** makes pairwise distances nearly identical, so distance stops being a useful measure of similarity and clustering breaks down.

High-dimensional spaces are also extremely **sparse**. Because volume grows exponentially with dimension, a fixed number of points becomes vanishingly sparse, and methods like DBSCAN need far more data to find dense neighborhoods.

Low Dimensions

2D scatter with varied pairwise distances



High Dimensions

Points appear almost equidistant; distances uniform

In low dimensions, distances vary and reveal structure. In high dimensions, they converge, removing the signal clustering methods need.

> Dimensionality reduction — PCA, autoencoders, or manifold methods — is often needed before distance-based unsupervised methods on high-dimensional data.

Anomaly Detection

APPLICATIONS

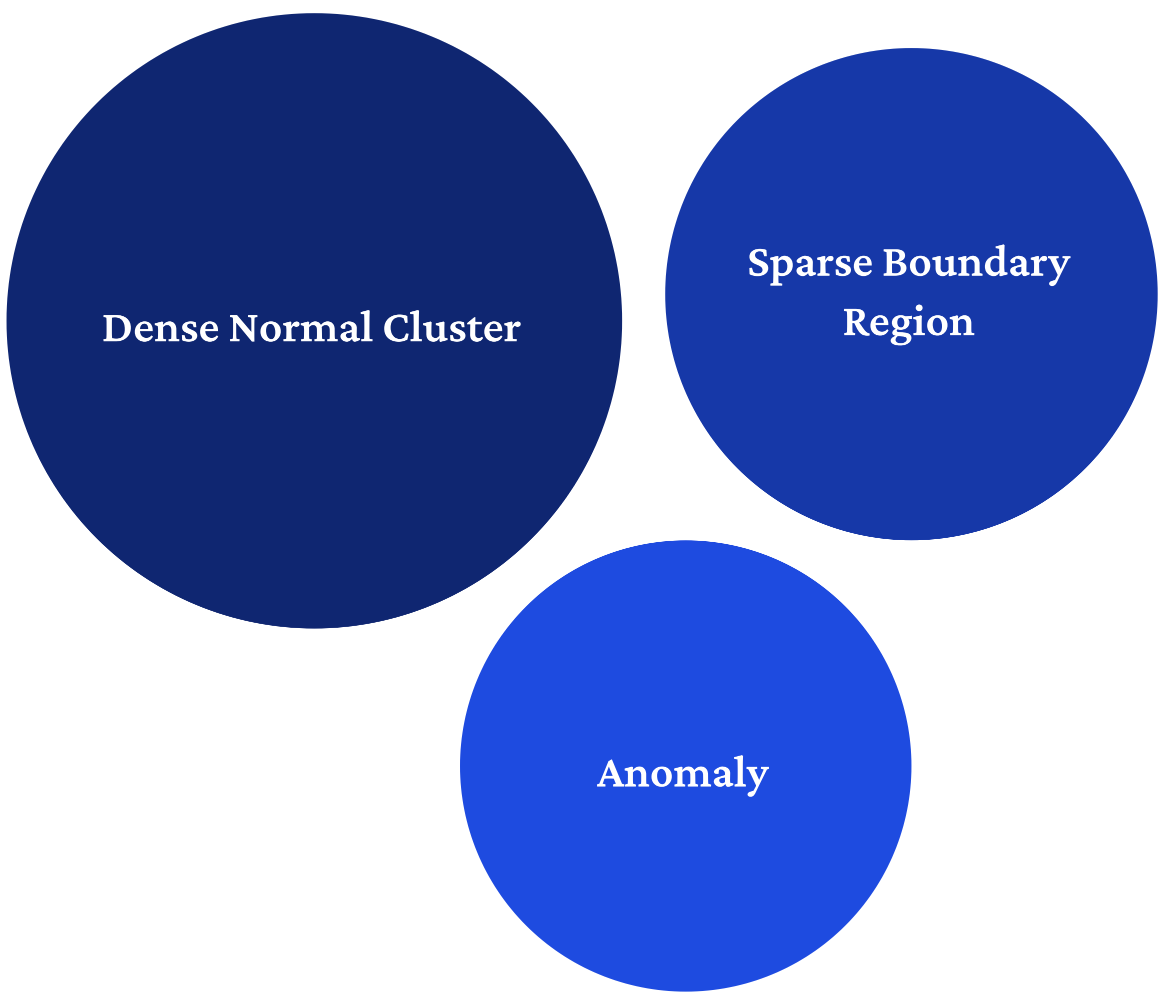
What Is an Anomaly?

An anomaly — or outlier — is an observation that differs sharply from the pattern learned from most of the data. In density-based terms, anomalies lie in low-probability regions of the estimated distribution.

Detection Approaches

Distance-based methods flag points far from their nearest neighbors. **Density-based** methods, including DBSCAN noise labels and Isolation Forest, target sparse regions. **Reconstruction-based** methods use autoencoders: high reconstruction error signals an anomaly.

The isolated point sits outside the normal density region. An unsupervised model gives it a low probability or high anomaly score — no labels needed.



Representation for Prediction

INTEGRATION

Modern machine learning rarely predicts directly from raw data. Instead, it uses a **representation learning** stage that turns inputs into a compact latent space before labels are applied. This pattern powers transfer learning, foundation models, and semi-supervised learning.

Raw inputs — pixels, text tokens, sensor readings — are often high-dimensional and noisy. A good representation removes redundancy, reduces noise, and exposes the latent factors that matter. The supervised stage then works on this cleaner encoding, often with far fewer labels.

> Unsupervised learning does not replace supervised learning — it strengthens it. Learned representations are the base of modern predictive systems.

The representation stage organizes the data without labels. The prediction stage then works in a cleaner space with fewer examples.

01

Raw Data

Noisy, high-dimensional inputs. Direct prediction is harder and more data-hungry.

02

Unsupervised Representation

Autoencoder, PCA, or contrastive learning creates a compact latent encoding without labels.

03

Supervised Prediction

A downstream model uses the clean latent space and generalizes better with fewer labels.